

RAGHUL RAJ NAVANEETHAKRISHNAN

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ML Systems · GPU-Accelerated Inference · Edge AI · Embedded Linux

TECHNICAL SKILLS

Languages	C++, C, Python, CUDA, OpenCL, Shell, BitBake
ML/AI	PyTorch, OpenCV, ONNX Runtime, TensorRT, scikit-learn, Pandas, Numpy
Optimization & Profiling	Custom CUDA/OpenCL kernels, quantization, Nsight Systems
Embedded & Systems	Yocto Project, Embedded Linux, ARMv8, OpenOCD, Jetson Nano, RPi, RTOS

WORK EXPERIENCE

GPU-Accelerated 3D Surrogate Model for Springback Prediction (Master Thesis)

Feb 2026 – Jul 2026

University of Stuttgart, Germany

- Built a DGCNN & Transformer model on point cloud samples to reproduce Ansys FEM workflows; achieved 50 ms inference per sample vs. 1.5-hour simulation runtime, a 100,000x throughput improvement.
- Implementing custom CUDA kernels for KNN graph construction and EdgeConv from scratch, reducing compute and memory overhead compared to PyTorch Geometric baselines.
- Applying IREE/MLIR compiler optimizations to the inference graph; trained on NVIDIA RTX 3090 with MLflow.

BSP Engineer, Working Student

Sep 2024 – Dec 2025

ETAS (Bosch Group), Abstatt, Germany

- Expanded the existing Linux BSP to integrate Podman container runtimes and BusyBox utilities using the Yocto Project; updated the BSP core layers from Kirkstone to Scarthgap.
- Integrated the Linux PREEMPT_RT (Real-Time) patch and benchmarked the performance using rt-test
- Developed an ARMv8 CPU tracer utilizing Program Counter sampling on a Raspberry Pi using JTAG and OpenOCD.
- Implemented API tracer tool to monitor system-level and external API calls across the Adaptive AUTOSAR VRTE.

AI Software Engineer, Working Student

Dec 2023 – Jun 2024

Bosch Mobility GmbH, Ludwigsburg, Germany

- Benchmarked LSTM, Transformer, 1D CNN, and encoder-decoder Seq2Seq architectures for synthetic ECU time-series generation, defining quantitative KPIs aligned to ISO 18571 automotive validation criteria.
- Designed a custom ISO 18571-aligned loss function replacing generic MSE, improving model validation score by 10% against the automotive-specific waveform correlation metric.
- Built a Flask-based multi-user GPU platform for on-premise model training, inference, and result visualization, consolidating team ML workflows onto shared hardware.

PROJECTS

Monocular Depth estimation on Edge Device - Deployed MiDaS DPT-Hybrid model using TensorRT; applied compilation, operator fusion and quantization to speed up end-to-end inference latency by 3x speedup. Deployed ONNX Runtime model on Raspberry Pi 5, achieved 15 FPS on CPU.

Inference Engine in C++/OpenCL - UNet inference engine in C++ with OpenCL without ML framework dependencies, writing GPU kernels for convolution, pooling, upsampling, and batch normalization individually. Achieved 600x speedup over single-threaded CPU baseline.

Book2Braille - Embedded AI solution to aid people with visual impairment to read equations and texts from scientific books by converting it into Nemeth-Braille, Built using OpenCV's edge-AI device OAK-D

EDUCATION

M.Sc. INFOTECH, Universität Stuttgart, Germany

Oct 2023 – Jul 2026 | GPA: 1.9

B.E. – Electrical and Electronics Engineering, Anna University, India

Aug 2015 – Apr 2019

EARLY CAREER EXPERIENCE

Senior Business Analyst, Infineon Technologies, Bengaluru, India

Feb 2021 – Sep 2023

Developed Python ETL pipelines and Tableau dashboards to automate KPI tracking across 1,000+ memory SKUs.

Graduate Trainee, Ashok Leyland, Chennai, India

Jul 2019 – Jan 2021

Built a Python framework to process Euro 6 and Euro 4 telematics sensor logs for performance and emission analysis.

CERTIFICATIONS

TinyML & Efficient Deep Learning – MIT OCW · Deep Learning Specialization - Coursera